

1))

compared to a normal two latch leader follower flip flop, adding a third latch changes when the output updates. The extra latch is clock enabled like the leader, so it only becomes transparent when the clock is high.

The follower latch is enabled when $CLK = 0$ because its enable is $\neg CLK$. That means the follower's internal node can change during the low phase.

The third latch is closed during $CLK = 0$, so those follower changes don't reach the output right away. When CLK goes to 1, the third latch opens and copies the follower's stored value to the output. After that, it holds the output steady for the rest of the cycle until the next time it opens.

So instead of the output moving during the low phase as a plain follower output would, the visible output updates at the rising edge and stays stable while the clock is low.

2b)) so let's let the three JK flip-flops be Q_2, Q_1, Q_0 . Input bit is X

first, the next-state equations (Q^+) I'm using for this specific 6-state encoding are:

$$Q_2^+ = (Q_0 \cdot \neg Q_1 \cdot \neg Q_2) + (Q_1 \cdot \neg Q_0 \cdot \neg X) + (X \cdot \neg Q_1 \cdot \neg Q_2)$$

$$Q_1^+ = (Q_1 \cdot X) + (Q_0 \cdot Q_2 \cdot \neg X) + (\neg Q_0 \cdot \neg Q_1 \cdot \neg Q_2 \cdot \neg X)$$

$$Q_0^+ = (Q_1 \cdot \neg Q_0) + (\neg Q_0 \cdot \neg X) + (Q_0 \cdot X \cdot \neg Q_1)$$

for JK, use:

$$J = \neg Q \cdot Q^+$$

$$K = Q \cdot \neg Q^+$$

so:

$$J_2 = (Q_0 \cdot \neg Q_1 \cdot \neg Q_2) + (X \cdot \neg Q_1 \cdot \neg Q_2) + (Q_1 \cdot \neg Q_0 \cdot \neg Q_2 \cdot \neg X)$$

$$K_2 = (Q_0 \cdot Q_2) + (Q_2 \cdot X) + (Q_2 \cdot \neg Q_1)$$

$$J_1 = (Q_0 \cdot Q_2 \cdot \neg Q_1 \cdot \neg X) + (\neg Q_0 \cdot \neg Q_1 \cdot \neg Q_2 \cdot \neg X)$$

$$K_1 = (Q_1 \cdot \neg Q_0 \cdot \neg X) + (Q_1 \cdot \neg Q_2 \cdot \neg X)$$

$$J_0 = (Q_1 \cdot \neg Q_0) + (\neg Q_0 \cdot \neg X)$$

$$K0 = (Q0 \cdot Q1) + (Q0 \cdot \neg X)$$

So output:

$$Y = (Q0 \cdot Q1 \cdot \neg X) + (Q1 \cdot X \cdot \neg Q0) + (Q2 \cdot X \cdot \neg Q0) + (\neg Q0 \cdot \neg Q1 \cdot \neg Q2 \cdot \neg X)$$

3)b)

same two memory FFs, plus one more DFF to delay the “hit”

$$Q1 = X(t-1)$$

$$Q0 = X(t-2)$$

Q2 = delayed output bit

so equations:

$$D1 = X$$

$$D0 = Q1$$

$$D2 = (\neg Q0) \cdot Q1 \cdot X$$

$$Y = Q2$$